

Python Programming

Comprehensive Theory, Core Concepts, Code Examples & 25 Practice Questions

1. Introduction to Python Programming

Python is a high-level, interpreted, dynamically typed, and object-oriented programming language designed by Guido van Rossum in 1991. It emphasizes code readability with its clean syntax and extensive standard libraries.

Key Features of Python

- **Easy to Learn & Read:** English-like syntax minimizing complex code structure.
- **Dynamically Typed:** Variable types are determined automatically at runtime.
- **Interpreted Language:** Executes code line-by-line using the Python Interpreter.
- **Cross-Platform & Open-Source:** Runs seamlessly on Windows, macOS, Linux, and Unix.
- **Rich Ecosystem:** Wide domain support (Data Science, Web Dev, AI/ML, Automation).

2. Core Python Concepts & Code Examples

1. Data Types & Variables

Python supports standard built-in data types: Numbers (int, float), Strings (str), Booleans (bool), and collection types (list, tuple, dict, set).

```
# Code Example
name = "Alice"      # String
age = 25            # Integer
is_active = True    # Boolean
scores = [90, 85]   # List

print(f"{name} is {age} years old.")
```

2. Control Flow Constructs

Conditional statements (if-elif-else) and loops (for, while) govern program execution flow based on logical evaluation.

```
# Code Example
num = 15
if num % 2 == 0:
    print("Even Number")
else:
    print("Odd Number")

for i in range(3):
    print("Loop Step:", i)
```

3. Data Structures: Lists, Tuples, Dictionaries & Sets

- **List:** Ordered, mutable collection ([1, 2, 3]).
- **Tuple:** Ordered, immutable collection ((1, 2, 3)).
- **Dictionary:** Unordered Key-Value pair collection ({ "a": 1 }).
- **Set:** Unordered collection of unique items ({1, 2, 3}).

```
# Code Example
student = {"name": "Bob", "grade": "A"}
student["age"] = 21
print(student["name"]) # Output: Bob
```

4. Functions & Scope

Functions are reusable blocks of code defined using the def keyword. They can accept positional, keyword, and variable-length arguments (*args, **kwargs).

```
# Code Example
def add_numbers(a, b=10):
    return a + b

result = add_numbers(5) # Output: 15
```

5. Object-Oriented Programming (OOP)

Python supports OOP concepts including Classes, Objects, Inheritance, Encapsulation, and Polymorphism.

```
# Code Example
class Animal:
    def __init__(self, name):
        self.name = name
    def speak(self):
        return f"{self.name} makes a sound."

dog = Animal("Buddy")
print(dog.speak())
```

3. Advanced Python Concepts

- **Exception Handling:** Catches runtime errors using try, except, finally, and else blocks.
- **List Comprehensions:** Concise syntax to construct lists: [x**2 for x in range(5)].
- **Modules & Packages:** Importing built-in or external libraries using import math or from os import path.
- **File I/O:** Reading and writing files securely using context managers: with open('file.txt', 'r') as f:..

4. Python Practice Questions (25 MCQs)

1. Who created the Python programming language?

- | | |
|-------------------|----------------------|
| A) Dennis Ritchie | B) Guido van Rossum |
| C) James Gosling | D) Bjarne Stroustrup |

2. Which key symbol is used to start a single-line comment in Python?

- | | |
|-------|-------|
| A) // | B) # |
| C) /* | D) -- |

3. What is the output of bool(0) in Python?

- | | |
|---------|----------|
| A) True | B) False |
| C) None | D) Error |

4. Which of the following data structures is immutable in Python?

- | | |
|---------------|----------|
| A) List | B) Tuple |
| C) Dictionary | D) Set |

5. What is the correct file extension for Python scripts?

- | | |
|------------|--------|
| A) .python | B) .py |
| C) .pyt | D) .pt |

6. What keyword is used to define a function in Python?

- | | |
|-------------|-----------|
| A) function | B) def |
| C) func | D) define |

7. What is the output of print(2 ** 3)?

- | | |
|------|------|
| A) 6 | B) 8 |
| C) 9 | D) 5 |

8. Which operator is used for integer (floor) division in Python?

- | | |
|------|--------|
| A) / | B) // |
| C) % | D) div |

9. Which built-in function returns the number of items in a list?

- | | |
|------------|-------------|
| A) count() | B) len() |
| C) size() | D) length() |

10. How do you start a class definition in Python?

- | | |
|-----------------------|-------------------------|
| A) class MyClass: | B) class MyClass() |
| C) def class MyClass: | D) create class MyClass |

11. What is the output of 'Python'[1:4]?

- | | |
|---------|--------|
| A) Pyt | B) yth |
| C) ytho | D) yt |

12. Which keyword is used to handle exceptions in Python?

- | | |
|----------|-----------|
| A) catch | B) except |
| C) error | D) handle |

13. What is the constructor method name in Python classes?

- A) `__init__()`
- B) `__construct__()`
- C) `init()`
- D) `new()`

14. Which method is used to add an element at the end of a list?

- A) `add()`
- B) `append()`
- C) `push()`
- D) `insert()`

15. What does the break statement do in a loop?

- A) Skips current iteration
- B) Exits the loop immediately
- C) Restarts the loop
- D) Pauses execution

16. What is the data type of the result of 10 / 2 in Python 3?

- A) `int`
- B) `float`
- C) `double`
- D) `decimal`

17. Which of the following creates an empty set in Python?

- A) `{}`
- B) `set()`
- C) `[]`
- D) `()`

18. What is an anonymous inline function in Python called?

- A) inline function
- B) lambda function
- C) dummy function
- D) macro function

19. Which statement is used to import modules into a Python file?

- A) `include`
- B) `import`
- C) `require`
- D) `load`

20. What is the output of `len({'a': 1, 'b': 2, 'c': 3})`?

- A) 6
- B) 3
- C) 2
- D) Error

21. What happens if you try to modify a tuple after creation?

- A) Tuple updates successfully
- B) Raises a `TypeError`
- C) Reverts to empty tuple
- D) Creates a new list

22. Which function converts a string to an integer in Python?

- A) `str()`
- B) `int()`
- C) `parse()`
- D) `float()`

23. What is the output of `print("Hello" + "World")`?

- A) Hello World
- B) HelloWorld
- C) Error
- D) None

24. What does the range(5) function generate?

- A) Numbers from 1 to 5
- B) Numbers from 0 to 4
- C) Numbers from 0 to 5
- D) 5 random numbers

25. Which block in exception handling runs regardless of whether an exception occurred or not?

- A) try
- B) finally
- C) except
- D) else

Answer Key

1. B	6. B	11. B	16. B	21. B
2. B	7. B	12. B	17. B	22. B
3. B	8. B	13. A	18. B	23. B
4. B	9. B	14. B	19. B	24. B
5. B	10. A	15. B	20. B	25. B